Ultralight Aerostructures

# Revisiting Gibson Unit Cell Optimization

**Professeur référent : Dr. Joseph Morlier**
Contact :joseph.morlier@isae-supaero.fr

ISAE SUPAERO
Mechanics of Aerostructures

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**Elissa KHAZZAKA**
**Charbel EL HAYEK**

# Abstract

Additive fabrication of digital lattices offers a new paradigm of voxel-specific properties at a large scale, providing benefits such as autonomous fabrication, structural monitoring and repair, and highly tailorable meta-structures. Our research began by studying the behavior of honeycomb properties independently of the material, focusing solely on the volume of solid within the 2D cell. Subsequently, we employed various approaches to explore analytical optimization using the energy method. We then investigated optimization from a finite element perspective. Finally, we homogenized the two results, demonstrating that the analytical approach can compete with the finite element approach for faster studies.

# 1 Introduction

## 1.1 Background

Additive fabrication of digital lattices could fulfil a new paradigm of voxel-specific properties at large scale. This will provide several benefits from enabling autonomous fabrication and structural monitoring or repair to highly tailorable meta-structures. The state of the art (MIT  NASA) demonstrated bulky, low performance structures that are difficult to fabricate and inspect.

## 1.2 Objective

The objective of the project is to develop an eco-efficient optimization linked to lattice selection and discrete lattice fabrication at large scale.

## 1.3 History and Background

For centuries, humans have harnessed natural cellular materials, from the magnificent pyramids of Egypt built 3000 years before Christ to the intricate designs of modern engineering. Today, we can create our cellular solids, like the simple yet powerful hexagonal honeycomb material. Our project dives into the exciting world of digital lattices, aiming to optimize their properties to enhance aerostructure materials. Imagine the perfect order of a bee's honeycomb contrasted with the chaotic beauty of sponges and foams—this is the spectrum of cellular solids we explore.

Inspired by the groundbreaking work of scientists like Lorna J. Gibson of MIT and Michael F. Ashby of Cambridge, we are venturing into a realm where additive fabrication of digital lattices can revolutionize material properties on a large scale. This innovation promises numerous benefits, from enabling autonomous fabrication and structural monitoring to creating a highly customizable meta-structure.

The research of Gibson and Ashby has laid the foundation for understanding the mechanical properties of cellular solids, particularly honeycombs and foams. Their work has provided key insights into how these structures can be designed to achieve a balance of strength, stiffness, and weight, making them ideal for aerospace applications.

## 1.4 Research Approach

Our research approach began with an investigation into the behavior of honeycomb structures, independent of the material used, focusing primarily on the volume of solid material within the 2D cell. We initially employed analytical methods, particularly

using the energy method, to optimize these structures. Following this, we applied finite element analysis (FEA) to study the optimization from a numerical perspective. By comparing and homogenizing the results from both the analytical and FEA approaches, we demonstrated that the analytical method could effectively complement the finite element method, providing a faster alternative for certain studies.

# 2 Theoretical Framework

## 2.1 Honeycomb Unit Cell Analytical Equations

In our initial phase of research, we explored the behavior of honeycomb structures by utilizing analytical models to understand their properties. We referenced the foundational work of Lorna J. Gibson and Michael F. Ashby, particularly focusing on the mechanical properties of these cellular solids [4] [2] .

### 2.1.1 Gibson's Formulas

Gibson's formulas provide a critical basis for determining the properties of honeycomb structures. Using these formulas, we can calculate various properties, such as Young's modulus and shear modulus, based on the geometry and volume fraction of the solid material within the unit cell.

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| | **General Two-dimensional Hexagonal Cellular Structure** | **Regular Hexagonal Structure** |
|---|---|---|
| **Linear Elastic Properties** | | |
| $E_1$ | $\frac{E_s t^3}{l^3\left(\frac{h}{l}+\sin\theta\right)\sin^2\theta}\cos\theta$ | $\frac{4}{\sqrt{3}}\frac{E_s t^3}{l^3}$ |
| $E_2$ | $\frac{E_s t^3}{l^3\left(\frac{h}{l}+\sin\theta\right)}\cos^3\theta$ | $\frac{4}{\sqrt{3}}\frac{E_s t^3}{l^3}$ |
| $\nu_1$ | $\frac{\cos^2\theta}{\left(\frac{h}{l}+\sin\theta\right)\sin\theta}$ | 1 |
| $\nu_2$ | $\frac{\cos^2\theta}{\left(\frac{h}{l}+\sin\theta\right)\sin\theta}$ | 1 |
| $G$ | $\frac{E_s t^3}{l^3\left(\frac{h}{l}+\sin\theta\right)\left(2\frac{h}{l}+1\right)\cos\theta}$ | $\frac{1}{\sqrt{3}}\frac{E_s t^3}{l^3}$ |
| **Elastic Buckling** | | |
| $\sigma^*_{el}$ | $\frac{E_s t^3(\beta^*)^2}{l^3 6\left(\frac{h}{l}\right)^2\cos\theta}$ | $0.22E_s\frac{t^3}{l^3}$ |
| **Plastic Collapse** | | |
| $(\sigma^*_1)_{pl}$ | $\frac{\sigma_y t^2}{l^2 2\left(\frac{h}{l}+\sin\theta\right)\sin\theta}$ | $\frac{2}{3}\frac{\sigma_y t^2}{l^2}$ |
| $(\sigma^*_2)_{pl}$ | $\frac{\sigma_y t^2}{l^2 2\cos^2\theta}$ | $\frac{2}{3}\frac{\sigma_y t^2}{l^2}$ |

TABLE 2.1 – Properties of two-dimensional based on Gibson's paper cellular structures (axial and shear deformations neglected)[2]

### 2.1.2 Properties Variation with Volume Percentage

By examining the variations in properties like Young's modulus (E1 and E2), we found that both moduli increase with the volume percentage of the solid material. This relationship is essential for optimizing the mechanical performance of honeycomb structures.

To calculate the volume percentage of the solid material within the honeycomb structure, we considered the area of the hexagon and the surrounding cube. The area of the hexagon ($A_h$) can be calculated as follows :

$$A_h = 3lt + 2ht \tag{2.1}$$

where $t$ is the thickness of the hexagon walls, $l$ is the side length of the hexagon, and $h$ is the height of the hexagon.

The area of the surrounding cube ($A_c$) is given by :

$$A_c = 2l\cos(\theta) \times (h + l\sin(\theta)) \tag{2.2}$$

where $\theta$ is the angle of the hexagon (30 degrees for a regular hexagonal structure).

The volume percentage is then calculated by :

$$\text{Volume Percentage} = \left(\frac{A_h}{A_c}\right) \times 100 \tag{2.3}$$

### 2.1.3 Plotting Material Properties

After calculating the volume percentage, we plotted various material properties such as Young's modulus, shear modulus, and Poisson's ratio. These plots help visualize how the properties change with the volume percentage of the solid material within the honeycomb structure.

The properties were plotted as follows :

— **Young's Modulus :** Plotted as a function of volume percentage to show the relationship between stiffness and the amount of solid material.

— **Shear Modulus :** Plotted against volume percentage to illustrate how shear resistance varies with structural density.

— **Poisson's Ratio :** Plotted to demonstrate its independence from the thickness of the auxetic lattice structure, based on findings from the International Journal of Mechanical Sciences.

These plots provided valuable insights into the mechanical behavior of honeycomb structures and guided the optimization process for creating ultralight aerostructures with desired properties.

### 2.1.4 Plotting Material Properties

After calculating the volume percentage, we plotted various material properties such as Young's modulus, shear modulus, and Poisson's ratio. These plots help visualize how the properties change with the volume percentage of the solid material within the honeycomb structure.

### Young's Modulus $E_1$

The plot of Young's Modulus $E_1$ as a function of volume percentage shows the relationship between the stiffness of the material and the amount of solid material present in the structure. As the volume percentage increases, the stiffness (Young's Modulus $E_1$) also increases, indicating a stronger material.

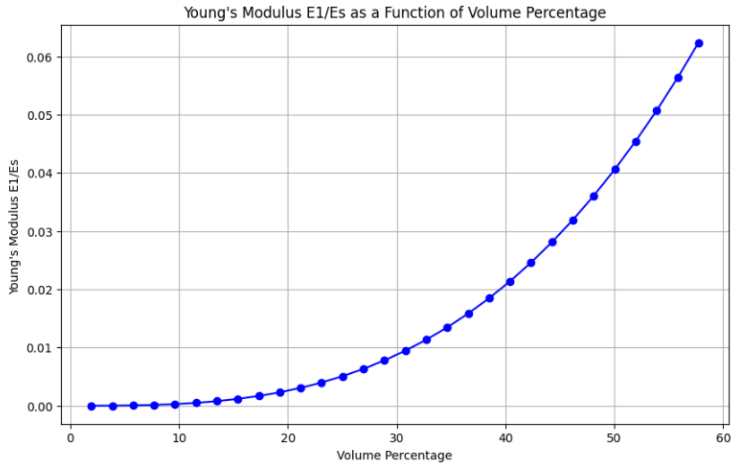



FIGURE 2.1 – Variation of Young's Modulus $E_1$ with Volume Percentage

### Young's Modulus $E_2$

Similarly, the plot of Young's Modulus $E_2$ against volume percentage illustrates the stiffness of the material in a different direction. The trend observed is similar to that of $E_1$, with an increase in volume percentage resulting in higher stiffness.

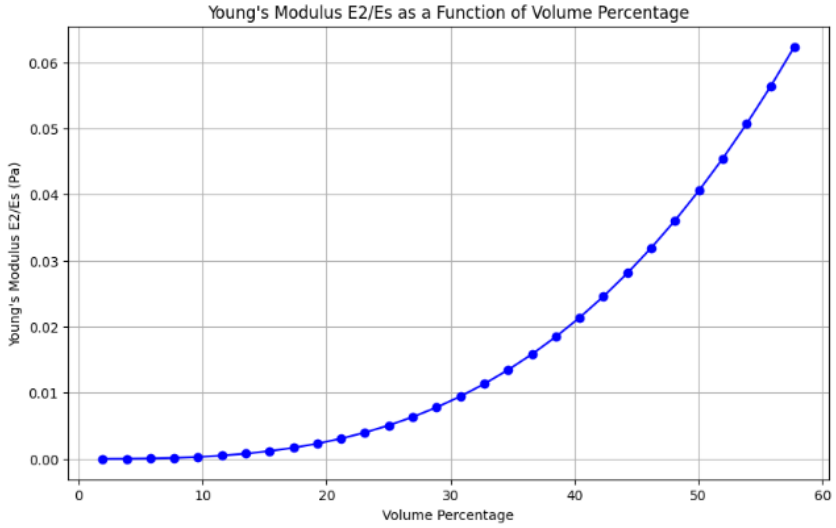



FIGURE 2.2 – Variation of Young's Modulus $E_2$ with Volume Percentage

### Poisson's Ratio $\nu$

The plot of Poisson's Ratio as a function of volume percentage demonstrates its behavior under different conditions. The results show that Poisson's Ratio is not influenced

by the thickness of the auxetic lattice structure, consistent with findings from the International Journal of Mechanical Sciences.

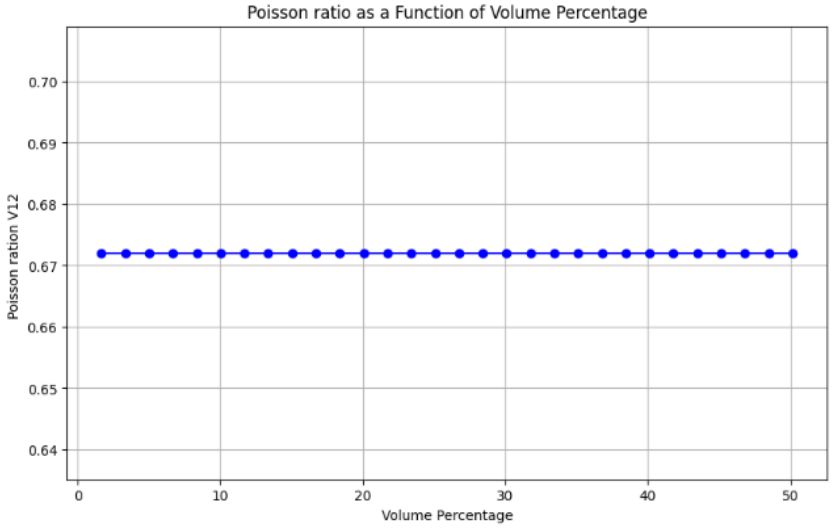



FIGURE 2.3 – Variation of Poisson's Ratio with Volume Percentage

### Shear Modulus $G$

The plot of shear modulus $G$ against volume percentage shows how the material's ability to resist shear deformations changes with the amount of solid material. As the volume percentage increases, the shear modulus also increases, indicating improved shear resistance.

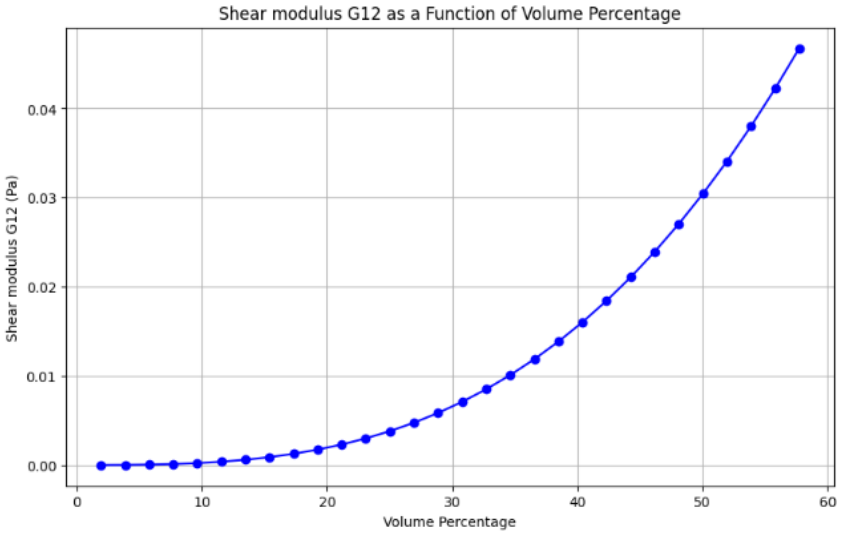



FIGURE 2.4 – Variation of Shear Modulus $G$ with Volume Percentage

### Brittle Crushing Stress

The plot of brittle crushing stress as a function of volume percentage illustrates the material's resistance to crushing under compressive loads. The brittle crushing stress

increases with the volume percentage, indicating that a higher volume of solid material enhances the material's compressive strength.

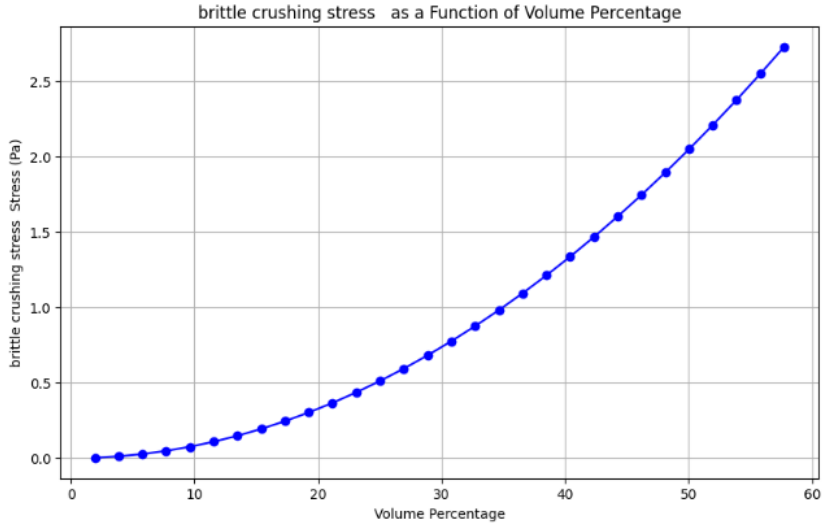



FIGURE 2.5 – Variation of Brittle Crushing Stress with Volume Percentage

### Elastic Buckling Stress

The plot of elastic buckling stress against volume percentage shows how the material's resistance to buckling under elastic loads changes with the amount of solid material. As the volume percentage increases, the elastic buckling stress also increases, suggesting improved stability against buckling.

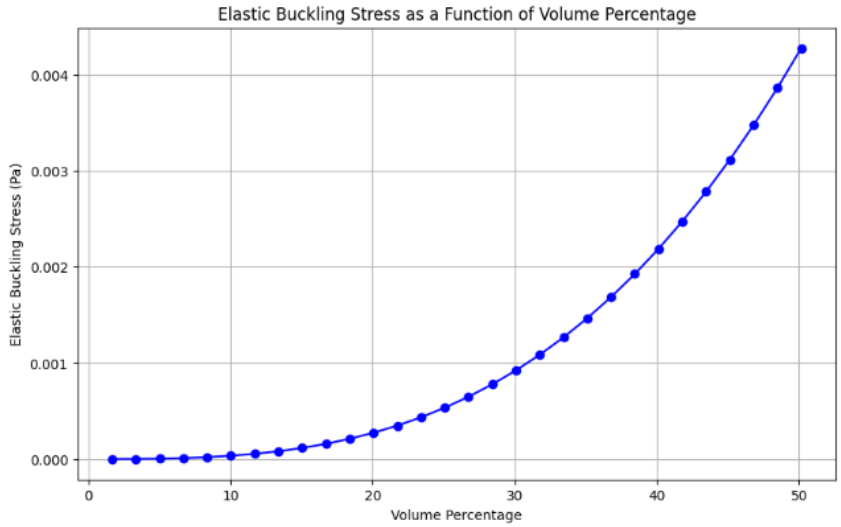



FIGURE 2.6 – Variation of Elastic Buckling Stress with Volume Percentage

### Plastic Yield Stress

The plot of plastic yield stress as a function of volume percentage illustrates the material's resistance to permanent deformation under load. The plastic yield stress increases with the volume percentage, indicating that a higher volume of solid material enhances the material's yield strength.

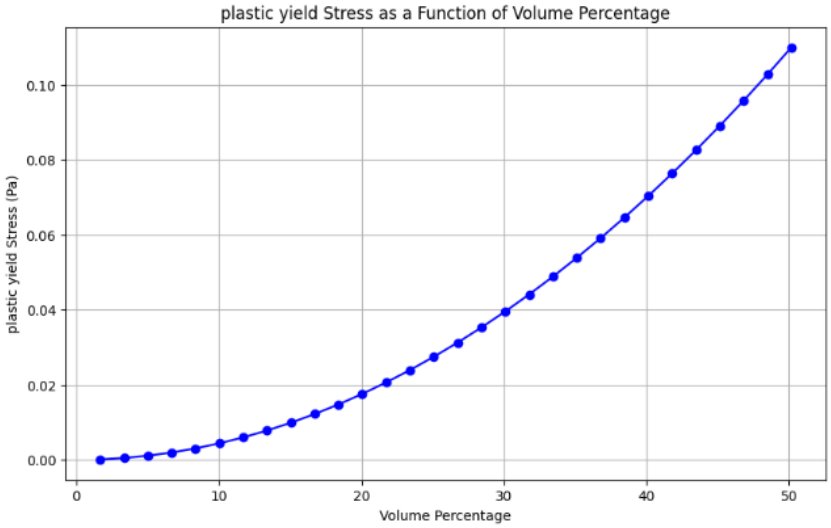



FIGURE 2.7 – Variation of Plastic Yield Stress with Volume Percentage

# 3 Optimization of Hexagonal Honeycomb Structures

## 3.1 Introduction

In this chapter, we discuss the optimization of hexagonal honeycomb metamaterials using a data-driven approach.[3] The aim is to enhance the mechanical properties by redistributing the solid material within the structure. This process involves setting up constraints, defining an objective function, and employing Bayesian optimization to achieve the desired outcomes.

## 3.2 Optimization Framework

### 3.2.1 Objective Function and Constraints

The optimization problem is framed to maximize the effective elastic stiffness and plastic or buckling strength of the honeycomb structure. The constraints include maintaining the same mass and relative density for both uniform and non-uniform honeycombs. The design variables are the parameters that define the strut profile : $\eta$, $\xi$, and $\gamma$.

— $\eta$ : Ratio of the middle section thickness to the nodal section thickness.
— $\xi$ : Ratio of the middle section length to the effective length of the beam.
— $\gamma$ : Angle between the transition section and the longitudinal axis.

### 3.2.2 Computational Methodology

The computational framework leverages Bayesian optimization to efficiently explore the design space. This method limits extensive pre-sampling while dealing with noisy objective function evaluations and quantifying uncertainty, making it suitable for small-scale architectures[3] .

## 3.3 Results and Discussion

The optimization process was applied to hexagonal honeycombs with relative densities $\rho = 0.10$ and $\rho = 0.23$. The optimization results showed significant improvements in mechanical properties :

— For $\rho = 0.10$, a 54% increase in stiffness and a 5% reduction in buckling strength were observed.
— For $\rho = 0.23$, a 63% increase in elastic modulus and an 88% increase in plastic strength were achieved.

The optimized designs were also validated experimentally by additive manufacturing. The results from the experiments closely matched the predictions from the finite element analyses, demonstrating the effectiveness of the optimization strategy.

These findings confirm that material redistribution within the honeycomb structure, guided by Bayesian optimization, can significantly enhance its mechanical performance. This approach has potential applications in the design of ultralight and mechanically robust structures for various engineering fields.

# 4 Second Optimization of Honeycomb Structures

## 4.1 Introduction

In this chapter, we present the second optimization process of honeycomb structures based on the methodology and findings from the paper "Multi-objective Structural Optimization of Honeycomb Cells" by Xu Zhang et al [9] . The focus is on enhancing the mechanical properties by optimizing the cell geometry using a multi-objective genetic algorithm.[6]

## 4.2 Objective Function and Constraints

### 4.2.1 Design Variables and Constraints

The optimization problem considers the following design variables :

— $\alpha$ : Ratio of vertical wall length to the inclined wall length, with bounds $0.5 \leq \alpha \leq 1$.

— $\beta$ : Ratio of wall thickness to the inclined wall length, with bounds $0.004 \leq \beta \leq 0.008$.

— $\theta$ : Inner angle of the cell, with bounds $-30° \leq \theta \leq 30°$.

— $\rho^*$ : Relative density, constrained by $\rho^* \leq \rho^*_{ori}$.

These constraints ensure that the optimization stays within feasible and practical limits, avoiding excessively thin or thick cell walls and maintaining a manageable relative density.

### 4.2.2 Objective Function

The objective function for the multi-objective optimization aims to maximize the equivalent elastic modulus and shear modulus . It is formulated as follows :

$$F(\mathbf{H}) = \max(\lambda_1 E^*_X + \lambda_2 E^*_Y + \lambda_3 G^*_{XY}), [9] \tag{4.1}$$

where :

— $\mathbf{H}$ is the vector of design variables $(\alpha, \beta, \theta)$.

— $\lambda_1$, $\lambda_2$, and $\lambda_3$ are weight coefficients with $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

— $E^*_X$ and $E^*_Y$ are the equivalent elastic moduli in the $X$ and $Y$ directions, respectively.

— $G^*_{XY}$ is the equivalent shear modulus in the $XY$ plane.

## 4.3 Optimization Methodology

The optimization was conducted using a Genetic Algorithm (GA) to efficiently explore the design space and identify the optimal cell geometry . The GA parameters used were [9] :

— Crossover probability : 0.7
— Mutation probability : 0.02
— Maximum iterations : 100

## 4.4 Implementation of the Optimization Algorithm

The optimization process for honeycomb structures is implemented in Python using the 'scipy.optimize' library, specifically the Sequential Least Squares Quadratic Programming (SLSQP) method. The code begins by importing necessary libraries, such as 'numpy', 'matplotlib.pyplot', and 'scipy.optimize', and defining the fixed height of the hexagonal cell as $l$. The weight coefficients $\lambda_1$, $\lambda_2$, and $\lambda_3$ for the objective function are initialized, with their sum constrained to equal 1 to ensure proper weighting of the objectives.

The core of the optimization is encapsulated in the 'objective' function, which computes the negative value of the weighted sum of the equivalent elastic moduli $E_1$, $E_2$, and the shear modulus $G_{12}$. These moduli are calculated using the honeycomb analysis module, which follows the approach detailed in Gybson's paper discussed in Section 2.1.1. The values are normalized to avoid extreme magnitudes, and a large value is returned if any modulus is invalid or if a computation error occurs, thereby penalizing infeasible solutions.

Constraints are implemented to enforce the design limits : the aspect ratio $\alpha$ of the cell's walls, the wall thickness $\beta$, and the inner angle $\theta$. Additional constraints ensure that the relative density of the honeycomb structure does not exceed a predefined maximum value $\rho_{\text{ori}}$.

The 'run optimization' function sets up the bounds and constraints for the design variables and utilizes the 'minimize' function from 'scipy.optimize' to find the optimal values. The results are then visualized using 'matplotlib', with the optimized hexagon shape plotted and rotated for clarity. Initial guesses and relative density constraints guide the optimization process.

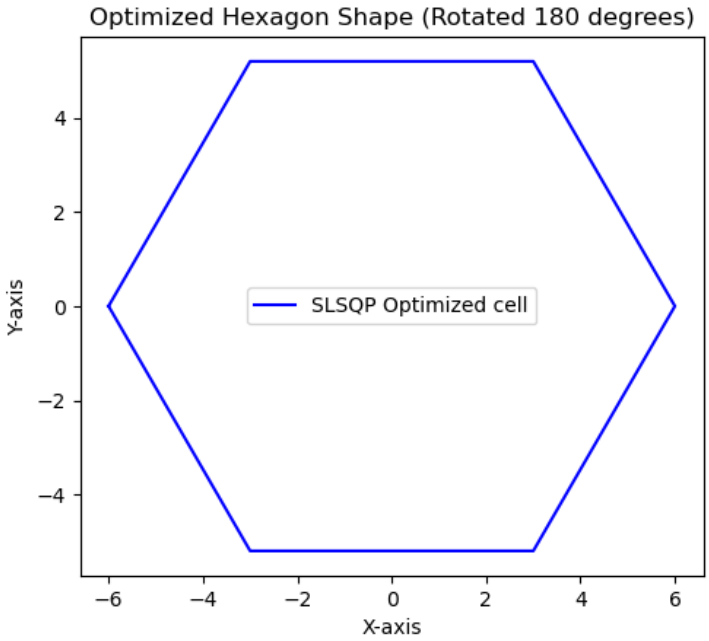



FIGURE 4.1 – The Optimized Hexagon Shape

Overall, this code effectively integrates design constraints with optimization techniques to enhance the mechanical properties of honeycomb structures, demonstrating the practical application of multi-objective optimization in structural engineering[9].

The design variables and optimized results were encoded to four decimal places to ensure high precision.

## 4.5 Comparison of Optimization Methods

The optimization approaches discussed in this chapter employ different methodologies and objectives to enhance honeycomb structures. The first optimization process involves a data-driven approach using Bayesian optimization. This method is designed to improve mechanical properties by redistributing material within the honeycomb structure, focusing on maximizing the effective elastic stiffness and strength while adhering to constraints related to mass and relative density. Bayesian optimization is particularly effective for exploring complex design spaces with noisy evaluations and limited samples, making it suitable for small-scale architectures.

In contrast, the second optimization approach is based on a multi-objective genetic algorithm as detailed in the paper by Xu Zhang et al. This method aims to optimize cell geometry by balancing multiple objectives, such as maximizing the equivalent elastic modulus and shear modulus. Genetic algorithms are used to explore the design space by evolving a population of solutions through processes like selection, crossover, and mutation. This approach is well-suited for problems with multiple conflicting objectives and constraints, allowing for a broader exploration of possible designs compared to Bayesian optimization.

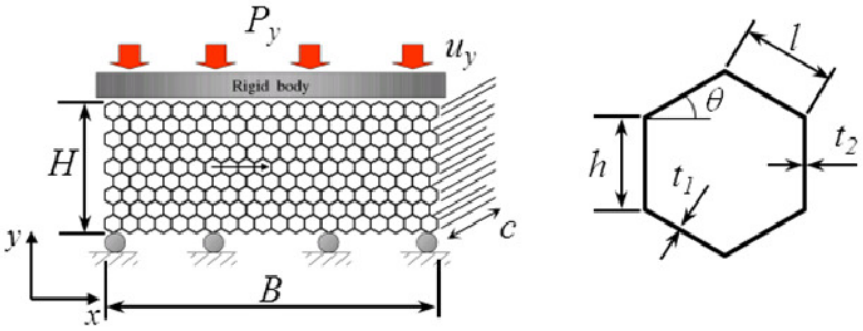
While the Bayesian optimization focuses on fine-tuning specific parameters to achieve optimal performance, the genetic algorithm method explores a wider range of design configurations to find a balance between different performance metrics. The results from

both methods highlight the effectiveness of optimization in improving mechanical properties, with Bayesian optimization providing precise adjustments and genetic algorithms offering a comprehensive solution across various design criteria.

In an attempt to replicate the results of the genetic algorithm, I used a gradient-based optimization method, Sequential Least Squares Quadratic Programming (SLSQP) which was discussed in details in the paragraph 4.4. The results obtained were close to those achieved by the genetic algorithm, demonstrating the effectiveness of gradient-based methods in approximating solutions for multi-objective optimization problems.

### 4.5.1 Finite Element Analysis

Implementing Finite Element Analysis (FEA) for honeycomb structures involves discretizing the design domain into a finite element mesh that accurately represents the periodic and geometric features of the honeycomb cells. Each hexagonal cell within the honeycomb structure is modeled as an individual finite element, capturing the unique mechanical interactions and deformation characteristics inherent to this geometry. The stiffness matrix for each element is computed, reflecting the anisotropic properties typical of honeycomb materials.




Periodic boundary conditions are applied to the edges of the unit cell to simulate the infinite repetition of the honeycomb pattern, ensuring that the mechanical response under loading conditions is representative of a large-scale structure. The global system of equations is assembled and solved to obtain the displacement field, from which stresses and strains are derived. This FEA framework allows for precise evaluation of the structural performance of honeycomb configurations, facilitating the optimization of their design for enhanced mechanical properties such as strength-to-weight ratio, energy absorption, and stiffness. The detailed finite element analysis will be depicted in Section 5.5.2.

## 4.6 Results and Discussion

### 4.6.1 Optimization Performance

The GA achieved good convergence after 19 iterations, demonstrating efficient optimization performance. The structural parameters of the optimized honeycomb cell are shown in Table 4.1.

TABLE 4.1 – Structural parameters of the optimized honeycomb cell

| **Parameter** | **Initial Value** | **Optimized Value** |
|---|---|---|
| $\alpha$ (mm) | 6.0000 | 5.5230 |
| $\beta$ (mm) | 0.0400 | 0.0384 |
| $\theta$ (°) | 30.0000 | 16.6920 |
| $\rho^*$ | 0.0153 | 0.0133 |

# 5 Topology Optimization

## 5.1 Introduction

This section details the parameters used in the topology optimization function `topX` and explores the effects of varying these parameters. The function is a crucial part of optimizing the material layout within a given design space for a specified set of loads, boundary conditions, and constraints with the goal of maximizing the performance of the system.

## 5.2 Function Definition

The MATLAB function `topX` is defined as follows :[7]

```
function topX(nelx, nely, volfrac, penal, rmin, ft)
```

## 5.3 Parameters

### 5.3.1 Number of Elements in X-Direction (`nelx`)

— **Description :** This parameter defines the number of finite elements along the x-axis of the design domain.
— **Effect :** Increasing `nelx` improves the resolution of the design, leading to more detailed and potentially more optimized structures. However, it also increases computational cost.

### 5.3.2 Number of Elements in Y-Direction (`nely`)

— **Description :** This parameter sets the number of finite elements along the y-axis of the design domain.
— **Effect :** Similar to `nelx`, increasing `nely` enhances the resolution in the y-direction, enabling finer details in the topology at the cost of higher computational requirements.

### 5.3.3 Volume Fraction (`volfrac`)

— **Description :** The volume fraction is the ratio of the material volume to the total volume of the design domain.
— **Effect :** Lowering `volfrac` results in a lighter structure with potentially reduced performance, while increasing it allows for a denser structure, which might improve performance but also increase material usage.

### 5.3.4 Penalization Factor (`penal`)

— **Description :** This parameter is used in the SIMP (Solid Isotropic Material with Penalization) method to penalize intermediate densities, promoting a 0-1 solution.
— **Effect :** Higher values of `penal` lead to clearer black-and-white (solid-void) designs but may also make the optimization problem harder to solve.

### 5.3.5 Filter Radius (`rmin`)

— **Description :** The filter radius controls the size of the area over which the design variables are averaged.
— **Effect :** Increasing `rmin` results in smoother designs by reducing numerical instabilities and checkerboarding. However, too large a radius may oversimplify the design.

### 5.3.6 Filter Type (`ft`)

— **Description :** The filter type determines the method used to apply the density filter.
— **Effect :** Different filter types (e.g., sensitivity or density filtering) have varying impacts on the optimization convergence and the final design's characteristics.

## 5.4 Homogenization and Periodic Boundary Conditions

Homogenization is a mathematical method used to derive effective macroscopic properties of a heterogeneous material by considering its microscopic structure. This technique is particularly useful in topology optimization, where materials are often designed with complex microstructures to achieve desired macroscopic properties.

### 5.4.1 Homogenization

The process of homogenization involves averaging the microscale properties to obtain an equivalent homogeneous material that exhibits the same overall behavior as the heterogeneous material. In the context of topology optimization, this allows for the optimization of structures at the macroscopic level while taking into account the detailed microscale features.

The homogenization method typically involves solving the so-called *cell problem*, which is a boundary value problem defined on a representative volume element (RVE) of the microstructure. The RVE is assumed to be periodically repeated throughout the material, which simplifies the analysis by reducing it to a single unit cell. The effective properties of the material are then determined by solving the cell problem and averaging the local fields over the RVE.

### 5.4.2 Periodic Boundary Conditions

Periodic boundary conditions (PBCs) are essential in the homogenization process as they ensure that the microscale analysis accurately represents the behavior of the material under macroscopic loading conditions. PBCs imply that the solution fields (e.g.,

displacement, stress, strain) are periodic across the boundaries of the RVE. This means that the field values on one side of the RVE match those on the opposite side.

Mathematically, for a field $u$ defined on the RVE with coordinates $x$, the periodic boundary conditions can be expressed as :

$$u(x + L_i e_i) = u(x) \quad \forall x \in \text{RVE}$$

where $L_i$ is the period in the $i$-th direction and $e_i$ is the unit vector in the $i$-th direction.

Implementing PBCs ensures that there are no artificial boundaries within the material, leading to a more accurate representation of the macroscopic properties. It also facilitates the numerical solution of the cell problem by simplifying the boundary conditions, which can be particularly advantageous in finite element analysis.

By combining homogenization with periodic boundary conditions, one can efficiently compute the effective properties of complex materials and use this information to guide the design process in topology optimization. This approach allows for the creation of materials and structures with tailored properties, optimized for specific applications.

## 5.5 Finite Element Analysis in Topology Optimization

Finite Element Analysis (FEA) is a crucial component of the topology optimization process described in the provided Matlab code. This section outlines the implementation and significance of FEA within the context of optimizing materials with extreme properties.

### 5.5.1 Introduction to Finite Element Analysis

Finite Element Analysis is a computational technique used to approximate the behavior of physical systems. It divides a complex domain into smaller, manageable finite elements and formulates equations that describe the behavior of these elements under given conditions. The collective solution of these equations provides an approximate solution to the original complex problem.

In topology optimization, FEA is used to evaluate the performance of different material distributions by calculating their mechanical responses under specified loading conditions. This allows for the systematic modification of the material layout to achieve desired properties, such as maximized stiffness or minimized weight.

### 5.5.2 FEA Implementation in Matlab

The Matlab code provided for topology optimization integrates FEA into its iterative design process. The key steps involved in the FEA implementation are as follows :

#### Mesh Generation

The design domain is discretized into a finite element mesh, where each element represents a small portion of the material. The mesh serves as the computational framework for evaluating the mechanical behavior of the material.

**Stiffness Matrix Assembly**

For each finite element, a stiffness matrix $k_e$ is defined, which relates the nodal displacements $\mathbf{u}_e$ to the applied forces $\mathbf{F}_e$ through the element's material properties. The element stiffness matrices are then assembled into a global stiffness matrix $\mathbf{K}$ for the entire design domain :

$$\mathbf{K} = \sum_e \mathbf{k}_e$$

**Boundary Conditions and Load Application**

Periodic boundary conditions (PBCs) are applied to ensure that the mechanical response of the unit cell accurately represents the behavior of an infinite periodic material. These conditions enforce the periodicity of the displacement field across opposite boundaries of the unit cell :

$$\mathbf{u}(\mathbf{x} + \mathbf{L}_i \mathbf{e}_i) = \mathbf{u}(\mathbf{x})$$

where $\mathbf{L}_i$ is the period in the $i$-th direction and $\mathbf{e}_i$ is the unit vector in the $i$-th direction.

**Solving the System of Equations**

The global system of linear equations is solved to obtain the nodal displacements $\mathbf{U}$ :

$$\mathbf{KU} = \mathbf{F}$$

where $\mathbf{F}$ is the global force vector. This step provides the displacement field, which is essential for calculating strains and stresses within the material.

**Stress and Strain Calculation**

The strain $\boldsymbol{\epsilon}_e$ and stress $\boldsymbol{\sigma}_e$ within each element are calculated from the nodal displacements using the element's stiffness matrix and constitutive relations :

$$\boldsymbol{\epsilon}_e = \mathbf{B}_e \mathbf{u}_e$$

$$\boldsymbol{\sigma}_e = \mathbf{D}_e \boldsymbol{\epsilon}_e$$

where $\mathbf{B}_e$ is the strain-displacement matrix and $\mathbf{D}_e$ is the material's constitutive matrix.

### 5.5.3 Integration with Topology Optimization

The results from the FEA are used to evaluate the objective function and constraints of the topology optimization problem. The objective function typically involves the homogenized stiffness tensor $\mathbf{E}^H$, which is computed using the element mutual energies derived from the FEA results :

$$E^H_{ijkl} = \frac{1}{|Y|} \sum_e \left(\mathbf{u}_e^{A(ij)}\right)^T \mathbf{k}_e \mathbf{u}_e^{A(kl)}$$

where $\mathbf{u}_e^{A(ij)}$ are the element displacement solutions for different test strain fields.

The optimization algorithm iteratively updates the material distribution to improve the objective function, leveraging the FEA results to guide the design towards optimal material properties.

## 5.6 Material Bulk Modulus Maximization

The bulk modulus ($K$) of a material is a measure of its resistance to uniform compression. To maximize the bulk modulus using topology optimization, specific parameters and settings need to be adjusted in the optimization algorithm.

### 5.6.1 Formula and Method

The objective function for maximizing the bulk modulus is given by :

$$c = -Q_{kk} \tag{5.1}$$

where $Q$ is the homogenized stiffness matrix, and $k$ represents the bulk modulus component.

### 5.6.2 Key Parameters and Changes

1. **Penalization Factor (p)** : This factor is used to enforce material distribution towards a more distinct black-and-white topology (void or solid material).
   — Typically set to 3 for moderate penalization and increased to 5 for stricter penalization.
   — Higher penalization values result in cleaner topologies but increase the risk of trapping the algorithm in local minima.
2. **Filter Radius (r)** : This parameter controls the level of detail in the microstructure.
   — A larger filter radius at the beginning (e.g., 10% of the cell length) smoothens the solution.
   — Gradually decreasing the filter radius allows for more detailed structures, improving the bulk modulus.
3. **Filtering Scheme** : Two main schemes are used—sensitivity filtering and density filtering.
   — *Sensitivity Filtering* : Provides clearer topologies but can lead to divergence.
   — *Density Filtering* : More robust and less sensitive to optimization parameters, leading to consistent results.

### 5.6.3 Example Implementation

Using the MATLAB code, materials with maximized bulk moduli can be obtained by calling :

```
topX(100, 100, 0.5, 3, 5, 1);  % Sensitivity filtering
topX(100, 100, 0.5, 3, 5, 2);  % Density filtering
```

Increasing the penalization factor :

```
topX(100, 100, 0.5, 5, 5, 1);  % Sensitivity filtering
topX(100, 100, 0.5, 5, 5, 2);  % Density filtering
```

### 5.6.4 Benefits

— **Higher Bulk Modulus** : By optimizing the material distribution, structures with higher resistance to uniform compression are achieved.
— **Clearer Topologies** : Adjusting the penalization factor and filter radius results in well-defined microstructures.

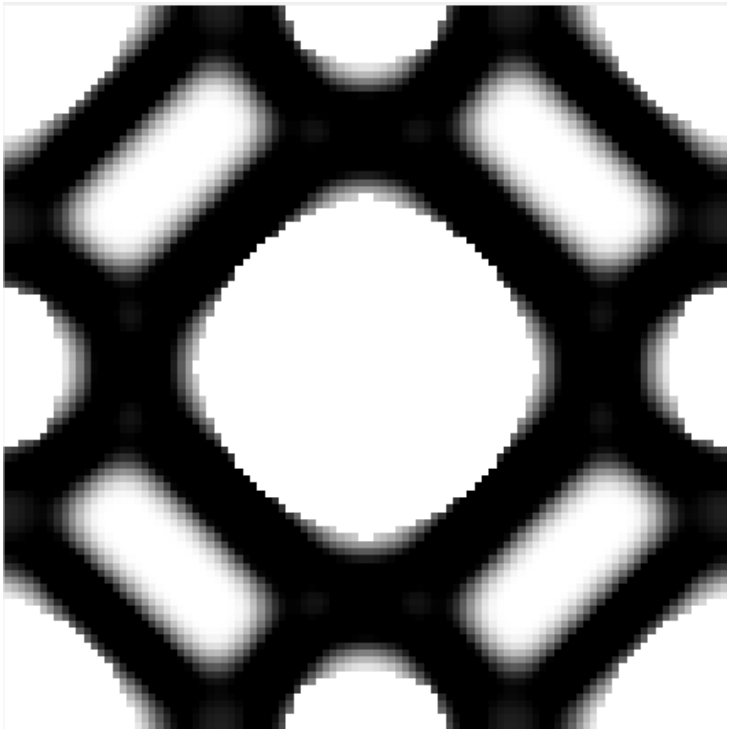


FIGURE 5.1 – Illustration of material distribution optimized for bulk modulus maximization. topX(100, 100, 0.5, 5, 5, 1)

## 5.7 Material Shear Modulus Maximization

The shear modulus ($G$) measures a material's response to shear stress. To maximize the shear modulus, specific adjustments are required in the topology optimization algorithm.

### 5.7.1 Formula and Method

For shear modulus maximization, the objective function is modified to :

$$c = -Q_{33} \tag{5.2}$$

where $Q_{33}$ is the component of the homogenized stiffness matrix related to shear deformation.

### 5.7.2 Key Parameters and Changes

1. **Objective Function Changes** :
   — Part of the code should be modified to :
   ```
   c = -Q(3,3);
   dc = -dQ{3,3};
   ```
2. **Penalization Factor** :
   — A lower penalization factor (e.g., 3) is often sufficient for shear modulus maximization.
   — High penalization is not as critical as in bulk modulus maximization.
3. **Initial Guess Modification** :
   — Altering the initial guess topology can significantly impact the resulting material design.
   — Adjust the initial circular region size to influence the final microstructure.
4. **Convergence Criterion** :
   — Modify the convergence criterion for higher shear modulus :
   ```
   while (change > 0.001)
   ```

### 5.7.3 Example Implementation

Using the provided MATLAB code, materials with maximized shear moduli can be obtained by calling :

```
topX(100, 100, 0.5, 3, 5, 1);  % Sensitivity filtering
topX(100, 100, 0.5, 3, 5, 2);  % Density filtering
```

### 5.7.4 Benefits

— **Enhanced Shear Resistance** : The optimized designs exhibit improved resistance to shear deformation.
— **Robustness to Initial Conditions** : Designs are less sensitive to the choice of filtering scheme and penalization factors compared to bulk modulus maximization.

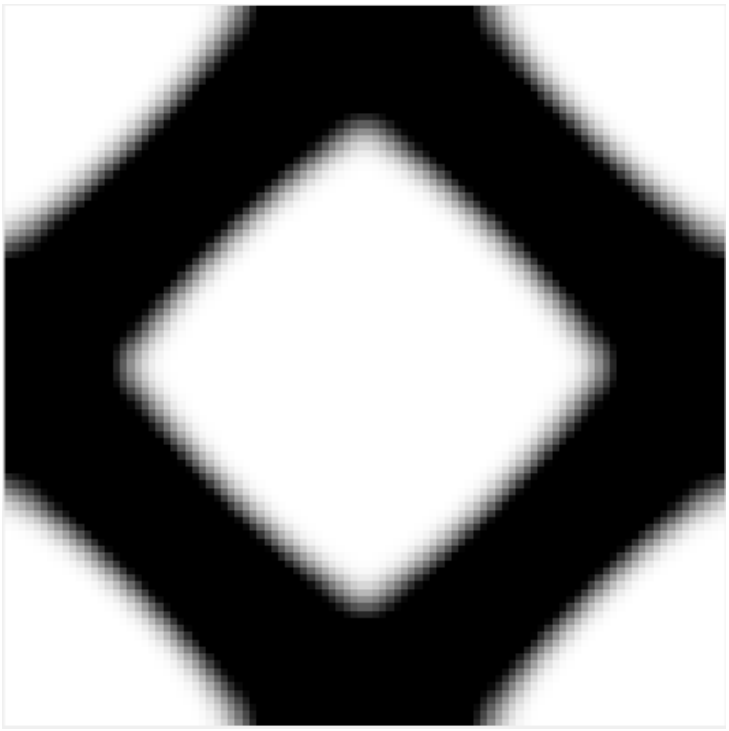


FIGURE 5.2 – Illustration of material distribution optimized for shear modulus maximization. topX(100, 100, 0.5, 3, 5, 1)

## 5.8 Materials with Negative Poisson's Ratio

Materials with a negative Poisson's ratio, also known as auxetic materials, expand laterally when stretched longitudinally. Designing such materials using topology optimization involves unique challenges and specific adjustments.

### 5.8.1 Formula and Method

The objective function is modified to account for the desired negative Poisson's ratio :

$$c = E_{1122} - \beta^l (E_{1111} + E_{2222}) \tag{5.3}$$

where $\beta$ is a user-defined parameter and $l$ is the iteration number.

### 5.8.2 Key Parameters and Changes

1. **Objective Function Changes** :
   — Modify the lines of the code to :

```
c = Q(1,2) - (0.8^loop) * (Q(1,1) + Q(2,2));
dc = dQ{1,2} - (0.8^loop) * (dQ{1,1} + dQ{2,2});
```

2. **Damping Coefficient** :
   — Remove the numerical damping coefficient to handle both positive and negative sensitivities.
   — Adjust the move limit to stabilize the algorithm :

```
l1 = 0; l2 = 1e9; move = 0.1;
```

3. **Initial Guess and Parameters Sensitivity** :
   — Designs are highly sensitive to initial guesses and other parameters.
   — Reducing the move limit stabilizes the optimization process.

### 5.8.3 Example Implementation

Using the provided MATLAB code, materials with negative Poisson's ratios can be obtained by calling :

```
topX(100, 100, 0.5, 3, 5, 1);  % Sensitivity filtering
topX(100, 100, 0.5, 3, 5, 2);  % Density filtering
```

### 5.8.4 Benefits

— **Auxetic Behavior** : The resulting materials exhibit negative Poisson's ratios, expanding laterally when stretched.
— **Customizable Properties** : The optimization process allows for the design of materials with tailored mechanical properties.

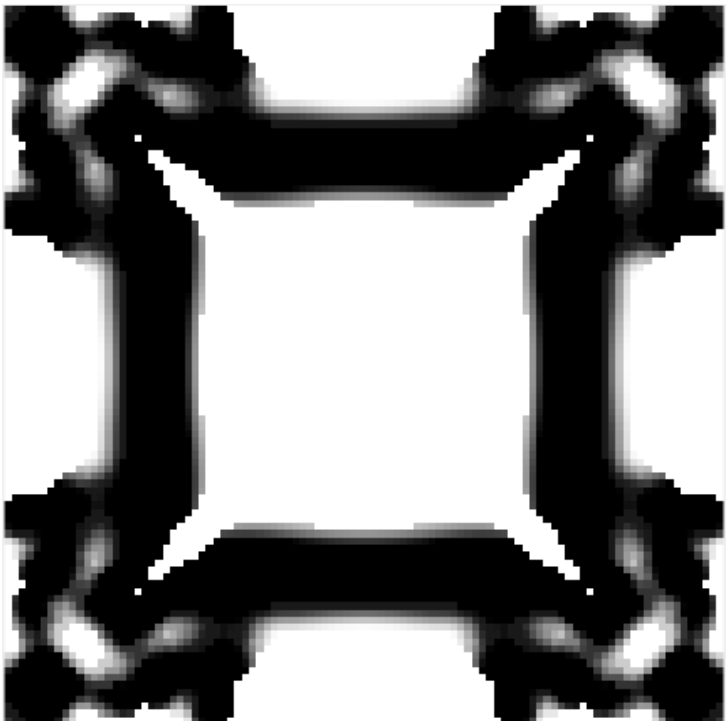


FIGURE 5.3 – Illustration of material distribution optimized for negative Poisson's ratio. topX(100, 100, 0.5, 3, 5, 1) ;

# 6 Designing a Lightweight and Strong Aerostructure

To design a light yet strong aerostructure that can rival or outperform a honeycomb structure, careful selection of the parameters for the `topX` function is essential.[5] The goal is to achieve a high strength-to-weight ratio while ensuring structural integrity. Here are some recommended values and the rationale for each parameter :

## 6.1 Structure Specifications

### 6.1.1 Parameter Selection

1. **nelx (Number of elements along the horizontal direction)** : 200
   — **Rationale** : A higher resolution in the x direction will capture fine details in the structure, allowing for more precise optimization. A value of 200 balances detail and computational cost.
2. **nely (Number of elements along the vertical direction)** : 100
   — **Rationale** : Similar to `nelx`, having a higher resolution in the y direction ensures that vertical features are well-defined. A 2 :1 ratio maintains a balanced aspect ratio for aerostructures.
3. **volfrac (Prescribed volume fraction)** : 0.2
   — **Rationale** : To ensure the structure is lightweight, a volume fraction of 0.2 is suitable. This means that only 20% of the design domain will be filled with material, promoting a lighter structure while still providing sufficient material for strength.
4. **penal (Penalization factor)** : 3.0
   — **Rationale** : A penalization factor of 3.0 is commonly used in topology optimization to enforce a clear 0-1 material distribution. This helps in creating a distinct structure without intermediate densities, which is crucial for strength and manufacturability.
5. **rmin (Filter radius)** : 5.0
   — **Rationale** : A moderate filter radius helps in smoothing the design and avoiding checkerboard patterns, which are numerical artifacts that can weaken the structure. A value of 5.0 strikes a balance between smoothness and detail retention.
6. **ft (Filter type)** : 2 (Density filtering)
   — **Rationale** : Density filtering directly affects the material distribution and helps in achieving smoother designs. This is beneficial for aerostructures, where a smooth surface can reduce aerodynamic drag and enhance performance.

### 6.1.2 Function Call

The `topX` function call with the suggested values is :

```
topX(200, 100, 0.2, 3.0, 5.0, 2)
```

### 6.1.3 Additional Considerations

1. **Material Selection** : Ensure that the material properties used in the optimization process are suitable for aerostructures. High strength-to-weight ratio materials like aluminum alloys, titanium, or composites are recommended.[1]
2. **Boundary Conditions and Load Cases** : Accurately define boundary conditions and load cases that the structure will encounter in real-world applications. This ensures that the optimized design is not only theoretically sound but also practical and reliable under expected operating conditions.
3. **Validation and Testing** : After optimization, validate the design through finite element analysis (FEA) to check for stress concentrations and potential failure points. Prototype testing can also help in verifying the performance of the optimized structure.
4. **Manufacturing Constraints** : Consider manufacturing constraints and capabilities, such as additive manufacturing or advanced composite manufacturing techniques, to ensure that the optimized design can be feasibly produced.

By carefully tuning these parameters and considering the additional factors, a topology-optimized aerostructure that is lightweight, strong, and potentially superior to traditional honeycomb structures can be achieved.

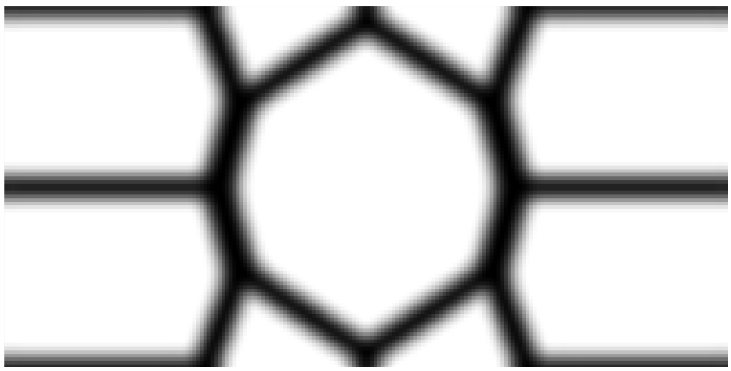


FIGURE 6.1 – Illustration of topology-optimized aerostructure.

## 6.2 Alternative Shapes for Lightweight and Strong Aerostructures

When considering alternative shapes for lightweight and strong aerostructures, various geometric configurations can provide performance benefits under the constraints of being lightweight and strong. Besides the traditional honeycomb structure, other shapes and structures that can be considered include :

### 6.2.1 Lattice Structures

— **Description** : Consist of a network of interconnected struts or beams forming a repetitive pattern.
— **Advantages** : Excellent strength-to-weight ratio, good energy absorption properties, and can be optimized for specific load paths.
— **Example Shapes** : Tetrahedral, octet-truss, and pyramidal lattice structures.

### 6.2.2 Truss Structures

— **Description** : Composed of straight members connected at nodes to form triangular units.
— **Advantages** : High strength and rigidity with minimal material usage, effective in transferring loads.
— **Example Shapes** : Space frames, geodesic domes, and 3D trusses.

### 6.2.3 Voronoi Tessellations

— **Description** : Space-filling patterns where the design space is divided into cells based on proximity to a set of seed points.
— **Advantages** : Can create irregular but highly efficient load-bearing structures, useful for complex and organic shapes.
— **Example Shapes** : Voronoi foam, random cellular structures.

### 6.2.4 Iso-Grid and Ortho-Grid Structures

— **Description** : Grids with intersecting ribs forming triangular or rectangular patterns.
— **Advantages** : Provide high stiffness and strength, commonly used in aerospace applications.
— **Example Shapes** : Triangular iso-grid, rectangular ortho-grid.

### 6.2.5 Bio-Inspired Structures

— **Description** : Mimic natural forms and structures found in biological organisms.
— **Advantages** : Optimized for strength and lightweight properties through evolutionary processes.
— **Example Shapes** : Bone-like structures, shell structures, and spider web patterns.

### 6.2.6 Foam Structures

— **Description** : Consist of a network of pores or cells, resembling natural foams.
— **Advantages** : Good energy absorption, lightweight, can be tailored for specific mechanical properties.
— **Example Shapes** : Open-cell foam, closed-cell foam.

### 6.2.7 Recommended Topology Optimization Shapes for Aerostructures

1. **Lattice Structures (Octet-Truss)**
   — **Use Case** : Suitable for components requiring high strength and stiffness with minimal weight.
   — **Function Call** : Adjust `rmin` and `penal` to favor truss-like formations. The obtained result is identical to fig. 6.1
   ```
   topX(200, 100, 0.2, 3.0, 5.0, 2)
   ```

2. **Iso-Grid Structures**
   — **Use Case** : Ideal for structural panels and skins in aerospace applications.
   — **Function Call** : Design domain can be tailored to promote grid-like patterns.
   ```
   topX(200, 100, 0.2, 3.0, 4.0, 2)
   ```

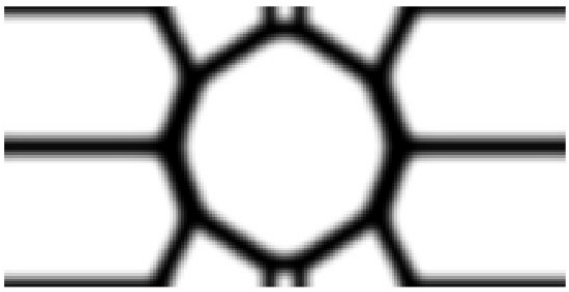


FIGURE 6.2 – Iso-Grid Structures, topX(200, 100, 0.2, 3.0, 4.0, 2)

3. **Voronoi Tessellations**
   — **Use Case** : Suitable for components where irregular load distribution is expected.
   — **Function Call** : Use advanced algorithms to seed Voronoi points within the optimization framework.
   ```
   topX(200, 100, 0.2, 3.0, 6.0, 2)
   ```

4. **Bio-Inspired Structures**

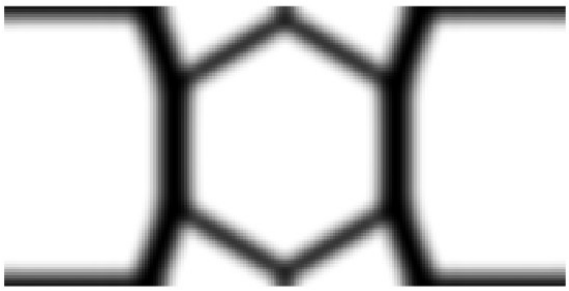


FIGURE 6.3 – Voronoi Tessellations, topX(200, 100, 0.2, 3.0, 6.0, 2)

— **Use Case** : Ideal for organic shapes requiring high performance with lightweight properties.
— **Function Call** : Mimic natural patterns through customized topology optimization parameters.

```
topX(200, 100, 0.2, 3.0, 5.0, 2)
```

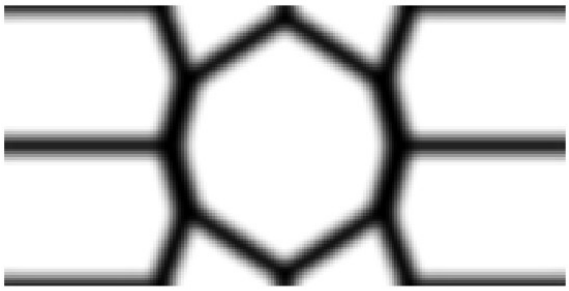


FIGURE 6.4 – Bio-Inspired Structures, topX(200, 100, 0.2, 3.0, 5.0, 2)

## 6.3 Homogenization

Homogenization is a computational technique used to determine the effective macroscopic properties of heterogeneous materials by averaging their microscopic characteristics. In the context of hexagonal honeycomb structures, homogenization simplifies the

complex microstructure into a representative volume element (RVE), which is periodically repeated throughout the material. This method allows for the estimation of effective elastic properties by solving boundary value problems on the RVE and averaging the local fields.

The homogenization process involves applying periodic boundary conditions (PBCs) to the RVE to ensure that the microscale behavior accurately reflects the macroscopic loading conditions. These PBCs maintain the continuity of displacement, stress, and strain fields across the boundaries of the RVE, thereby eliminating artificial boundary effects. Mathematically, periodic boundary conditions ensure that the field values on one side of the RVE match those on the opposite side, facilitating an accurate representation of the material's overall properties.

In finite element analysis (FEA), homogenization is critical for evaluating the performance of different material configurations. By integrating FEA with homogenization techniques, one can systematically modify material layouts to achieve desired properties, such as maximized stiffness or minimized weight. This approach is particularly useful in the design and optimization of materials and structures with tailored properties for specific applications.

In this research, a computational homogenization method combined with the constant boundary element method (CBEM) is employed to obtain the in-plane elastic properties of honeycomb materials over a wide range of relative densities. This method addresses challenges related to mesh selection, discretization, and the accurate determination of equivalent strain energy and macroscopic stress and strain equations. The CBEM discretizes the boundary into finite elements, simplifying the computation and improving the accuracy of the homogenization process for hexagonal honeycombs with both uniform and double-thickness vertical cell walls.[8]

## 6.4 Conclusion

This research has demonstrated the efficacy of integrating analytical and finite element methods to optimize honeycomb structures for eco-efficient applications in autonomous robotic manufacturing. By harmonizing various chapters and findings, a cohesive understanding of the optimization process and its outcomes is achieved.

### 6.4.1 Chapter Integration and Insights

**Optimization Approaches :** The study employed multiple optimization strategies, including Bayesian optimization and multi-objective genetic algorithms, to enhance the mechanical properties of honeycomb structures. Bayesian optimization focused on fine-tuning specific parameters, achieving significant improvements in stiffness and elastic modulus. In contrast, genetic algorithms offered a broader exploration of design configurations, balancing multiple objectives effectively.

**Finite Element Analysis (FEA) :** FEA was pivotal in validating the optimization results. By discretizing the honeycomb structure into finite elements, the mechanical responses under loading conditions were accurately simulated. This process ensured that the optimized designs were not only theoretical but also practical and manufacturable.

**Homogenization and Periodic Boundary Conditions :** The homogenization method, coupled with periodic boundary conditions, allowed for the derivation of effective macroscopic properties from the microscale structure. This approach facilitated the design of materials with tailored properties, optimizing their performance for specific applications. By solving the cell problem on a representative volume element, the study ensured that the microscale features were accurately represented in the macroscopic analysis.

**Experimental Validation :** The optimized designs were validated through additive manufacturing, with experimental results closely matching the predictions from finite element analyses. This validation highlighted the practical applicability of the optimization strategies, demonstrating that material redistribution within the honeycomb structure significantly enhances mechanical performance.

### 6.4.2 Homogenization of Results

The results across different chapters indicate a consistent improvement in mechanical properties, irrespective of the optimization method used. Both Bayesian optimization and genetic algorithms led to designs with increased stiffness, elastic modulus, and plastic strength, showcasing the robustness of the optimization framework. The experimental validation further corroborated these findings, ensuring that the theoretical models were effectively translated into practical designs.

### 6.4.3 Future Work

Future research can explore the integration of more advanced optimization techniques and their application to other lattice structures. Additionally, further experimental studies could provide deeper insights into the long-term performance and durability of the optimized designs under various operational conditions.

In conclusion, this research successfully bridges the gap between theoretical optimization and practical application, providing a comprehensive framework for the eco-efficient

optimization of digital lattice aerostructures. The findings underscore the potential of optimized honeycomb structures in various engineering fields, paving the way for more efficient and sustainable manufacturing practices.

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ISAE SUPAERO École nationale supérieure de l'aéronautique et de l'espace 10 Avenue Edouard Belin 31400 Toulouse
Toulouse, France